

Practice Question Sets

Optional Computer Science — Class IX — First Terminal Examination 2083

Sub code '3031' • Full Marks: 50 • Time: 2 hrs (per set)

Symbol No. _____

Sub code '3031'

Practice Question — Set 1

First Terminal Examination - 2083

Subject: Optional Computer Science

Class: IX Group 'A'

Full Marks: 50 Time: 2 hrs

Group 'A'

Rewrite the correct option in your answer sheet. [10 × 1 = 10]

1. Which fraction of a second is equivalent to one millionth of a second?

- (a) Millisecond (b) Microsecond (c) Nanosecond (d) Picosecond

2. What is the primary function of the CPU in a computer system?

- (a) Memory storage (b) Data input (c) Arithmetic and logic operations (d) Graphics processing

3. What does the term "BIOS" stand for in a computer system?

- (a) Basic Input Output System (b) Binary Input Output System (c) Base Input Output Software (d) Basic Integrated Operating System

4. In the hexadecimal number system, what does the symbol 'C' represent?

- (a) 11 (b) 12 (c) 10 (d) 13

5. The _____ tag in HTML makes your text, image, graphics, and videos move or scroll on your webpage.

- (a) <MARQUEE> (b)
 (c) <FORM> (d)

6. What is the correct HTML tag for inserting a horizontal line?

- (a)
 (b) <hr> (c) <href> (d) <td>

7. Which type of block in Scratch is used to repeat actions?

- (a) Motion Block (b) Control Block (c) Looks Block (d) Sensing Block

8. There are _____ extensions available in Scratch programming.

- (a) 5 (b) 9 (c) 11 (d) 12

9. The main purpose of the internet is to.....

- (a) Send emails only (b) Play games (c) Edit images (d) Access and display hypertext documents on the web

10. What are individual entries in a blog called?

- (a) Pages (b) Posts (c) Sections (d) Articles

Group 'B'

11. Write short answers to the following questions. [10 × 2 = 20]

(a) Why is a computer called an accurate machine?

(b) What is the role of the motherboard in a computer system?

(c) How does a peripheral device enhance a computer system?

(d) What is the hexadecimal number system? Write down the symbols used in the hexadecimal number system.

(e) Write any two advantages of block-based programming.

(f) Write any four application areas of Arduino UNO.

(g) Explain any four components of Scratch.

(h) Draw/list the stages of WDLC (Website Development Life Cycle).

(i) What are the uses of a search engine? Write any two popular search engines.

(j) Write down the uses of social media with its examples.

Group 'C'

12. Write long answers to the following questions. [4 × 5 = 20]

- a) Differentiate between SRAM and DRAM. [4]
- b) Define blog. Write down the pros and cons of blogging. [4]
- c) Calculate the following as indicated: [4]
 - i) $(111001)_2 + (100111)_2$
 - ii) $(100110)_2 - (10101)_2$
 - iii) $(F00D)_{16}$ into $(?)_2$
 - iv) $(615)_{10}$ into $(?)_8$
- d) Write an HTML program to create a web page including the following tags: [4]
 - i) Background color
 - ii) Hyperlink
 - iii) Marquee
 - iv) Radio button
- e) Define micro:bit with its application. [4]

— *Best of Luck* —

Solutions — Practice Set 1

Group 'A' — Answer Key

1. Which fraction of a second is equivalent to one millionth of a second? → **(b) Microsecond**
2. What is the primary function of the CPU in a computer system? → **(c) Arithmetic and logic operations**
3. What does the term "BIOS" stand for in a computer system? → **(a) Basic Input Output System**
4. In the hexadecimal number system, what does the symbol 'C' represent? → **(b) 12**
5. The _____ tag in HTML makes your text, image, graphics, and videos move or scroll on your webpage. → **(a) <MARQUEE>**
6. What is the correct HTML tag for inserting a horizontal line? → **(b) <hr>**
7. Which type of block in Scratch is used to repeat actions? → **(b) Control Block**
8. There are _____ extensions available in Scratch programming. → **(c) 11**
9. The main purpose of the internet is to..... → **(d) Access and display hypertext documents on the web**
10. What are individual entries in a blog called? → **(b) Posts**

Group 'B' — Answers

- (a)** A computer is called an accurate machine because it performs calculations and processes data with a very high degree of precision, producing errorless results as long as correct data and instructions are given (Garbage In, Garbage Out).
- (b)** The motherboard is the main circuit board of a computer that connects and allows communication between the CPU, RAM, storage devices, and other hardware components, acting as the backbone of the system.
- (c)** Peripheral devices such as a keyboard, mouse, printer, and scanner allow users to input data into and receive output from the computer, extending its functionality and making it more interactive and useful.
- (d)** The hexadecimal number system is a base-16 number system that uses sixteen symbols to represent values. The symbols used are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F (where A=10, B=11, C=12, D=13, E=14, F=15).
- (e)** 1) It is easy for beginners to learn since blocks are dragged and dropped instead of typing syntax. 2) It reduces syntax errors, allowing learners to focus on logic-building rather than memorizing commands.
- (f)** 1) Home automation systems 2) Robotics projects 3) Weather monitoring stations 4) Automatic plant watering / agriculture systems.
- (g)** 1) Stage — the area where the sprite's actions are displayed. 2) Sprite — a character or object that is programmed to perform actions. 3) Script area — where blocks are dragged and snapped together to build a program. 4) Block palette — the panel containing categories of blocks (Motion, Looks, Sound, etc.) used to build scripts.
- (h)** The stages of WDLC are: 1) Planning 2) Analysis 3) Design 4) Development 5) Testing 6) Deployment 7) Maintenance.
- (i)** A search engine helps users find information, websites, images, and videos on the internet quickly by entering keywords. Two popular search engines are Google and Bing.
- (j)** Social media is used for communicating with friends and family, sharing information and opinions, marketing businesses, and staying updated with news and trends. Examples: Facebook and Instagram.

Group 'C' — Answers

(a)

1) SRAM (Static RAM) does not need to be refreshed periodically, while DRAM (Dynamic RAM) needs to be refreshed thousands of times per second to retain data. 2) SRAM is faster but more expensive, while DRAM is slower but cheaper. 3) SRAM uses flip-flops (transistors) to store each bit, while DRAM uses a capacitor and a transistor. 4) SRAM is generally used for cache memory, while DRAM is used as main memory (RAM) in computers.

(b)

A blog is a regularly updated website or webpage, typically run by an individual or small group, where entries (posts) are written in an informal or conversational style and displayed in reverse chronological order.

Pros: 1) Easy way to share knowledge and opinions with a wide audience. 2) Can help build a personal brand or generate income through advertisements.

Cons: 1) Requires continuous time and effort to keep updated. 2) Content can be copied or misused by others, and maintaining quality/originality can be challenging.

(c)

i) $111001 = 57$, $100111 = 39$. Sum = $57 + 39 = 96 \rightarrow (1100000)_2$

ii) $100110 = 38$, $10101 = 21$. Difference = $38 - 21 = 17 \rightarrow (10001)_2$

iii) $F00D \rightarrow F=1111, 0=0000, 0=0000, D=1101 \rightarrow (1111000000001101)_2$

iv) $615 \div 8 = 76 \text{ rem } 7$; $76 \div 8 = 9 \text{ rem } 4$; $9 \div 8 = 1 \text{ rem } 1$; $1 \div 8 = 0 \text{ rem } 1 \rightarrow$ reading remainders bottom-up $\rightarrow (1147)_8$

(d)

```
<html>
<body bgcolor="lightblue">
  <h2>My Practice Web Page</h2>
  <a href="https://www.google.com">Visit Google</a><br><br>
  <marquee>Welcome to my HTML practice page!</marquee><br>
  <p>Select your favourite subject:</p>
  <input type="radio" name="subject" value="Computer"> Computer<br>
  <input type="radio" name="subject" value="Science"> Science<br>
</body>
</html>
```

(e)

The micro:bit is a small, pocket-sized programmable microcontroller board developed by the BBC, equipped with an LED display, buttons, sensors, and Bluetooth, designed to teach coding and electronics to students in a fun, hands-on way.

Applications: It is used to build simple electronic projects such as digital dice, step counters, temperature monitors, and simple robots, and is widely used in schools to teach programming and physical computing.

Symbol No. _____

Sub code '3031'

Practice Question — Set 2

First Terminal Examination - 2083

Subject: Optional Computer Science

Class: IX Group 'A'

Full Marks: 50 Time: 2 hrs

Group 'A'

Rewrite the correct option in your answer sheet. [10 × 1 = 10]

1. Which unit of time is equal to one billionth of a second?

- (a) Millisecond (b) Microsecond (c) Nanosecond (d) Picosecond

2. Which component of a computer is known as its "brain"?

- (a) RAM (b) CPU (c) Hard disk (d) Monitor

3. What is the function of BIOS when a computer is switched on?

- (a) It stores user files permanently (b) It performs POST and helps load the operating system (c) It connects the computer to the internet (d) It compresses data files

4. In the hexadecimal number system, what does the symbol 'A' represent?

- (a) 9 (b) 10 (c) 11 (d) 16

5. Which HTML tag is used to create a hyperlink?

- (a) <link> (b) <a> (c) <href> (d) <hlink>

6. Which HTML tag is used to display an image on a webpage?

- (a) (b) <src> (c) <pic> (d) <image>

7. Which type of block in Scratch is used to change a sprite's appearance (e.g., say, costume)?

- (a) Motion Block (b) Control Block (c) Looks Block (d) Sound Block

8. Scratch programming primarily uses which type of programming approach?

- (a) Text-based (b) Block-based (c) Machine-level (d) Assembly-based

9. Which of the following is an example of a web browser?

- (a) MS Word (b) Google Chrome (c) Adobe Photoshop (d) Windows Media Player

10. What is the regularly updated section/entry of a blog called?

- (a) Widget (b) Post (c) Template (d) Theme

Group 'B'

11. Write short answers to the following questions. [10 × 2 = 20]

(a) Why is a computer called a versatile machine?

(b) What is the role of RAM in a computer system?

(c) How does an input device differ from an output device? Give one example each.

(d) What is the octal number system? Write down the symbols used in the octal number system.

(e) Write any two advantages of using Scratch for beginners.

(f) Write any four application areas of micro:bit.

(g) Explain any four blocks/categories available in Scratch.

(h) List the stages of SDLC (Software/Website Development Life Cycle).

(i) What is a web browser? Write any two popular web browsers.

(j) Write down the advantages of e-mail with examples of any two email service providers.

Group 'C'

12. Write long answers to the following questions. [4 × 5 = 20]

- a) Differentiate between primary memory and secondary memory. [4]
- b) Define search engine. Write down its pros and cons. [4]
- c) Calculate the following as indicated: [4]
 - i) $(110101)_2 + (101011)_2$
 - ii) $(111010)_2 - (011011)_2$
 - iii) $(2A3)_{16}$ into $(?)_2$
 - iv) $(450)_{10}$ into $(?)_8$
- d) Write an HTML program to create a web page including the following tags: [4]
 - i) Font color
 - ii) Image insertion
 - iii) Ordered list
 - iv) Checkbox
- e) Define Arduino UNO with its features/applications. [4]

— **Best of Luck** —

Solutions — Practice Set 2

Group 'A' — Answer Key

1. Which unit of time is equal to one billionth of a second? → (c) **Nanosecond**
2. Which component of a computer is known as its "brain"? → (b) **CPU**
3. What is the function of BIOS when a computer is switched on? → (b) **It performs POST and helps load the operating system**
4. In the hexadecimal number system, what does the symbol 'A' represent? → (b) **10**
5. Which HTML tag is used to create a hyperlink? → (b) **<a>**
6. Which HTML tag is used to display an image on a webpage? → (a) ****
7. Which type of block in Scratch is used to change a sprite's appearance (e.g., say, costume)? → (c) **Looks Block**
8. Scratch programming primarily uses which type of programming approach? → (b) **Block-based**
9. Which of the following is an example of a web browser? → (b) **Google Chrome**
10. What is the regularly updated section/entry of a blog called? → (b) **Post**

Group 'B' — Answers

- (a) A computer is called a versatile machine because it can perform many different types of tasks — such as calculations, drawing, writing, playing media, and communication — simply by running different software programs.
- (b) RAM (Random Access Memory) temporarily stores data and instructions that the CPU needs while a program is running, allowing for fast read/write access; its contents are lost when the computer is powered off.
- (c) An input device is used to enter data and instructions into a computer, e.g., a keyboard, whereas an output device is used to display or present processed results from the computer, e.g., a monitor.
- (d) The octal number system is a base-8 number system that uses eight symbols to represent values: 0, 1, 2, 3, 4, 5, 6, 7.
- (e) 1) It has a simple, colourful drag-and-drop interface that makes learning to code fun and engaging. 2) It helps beginners understand programming logic (loops, conditions, variables) without worrying about complex syntax.
- (f) 1) Building simple games and animations 2) Creating step counters or pedometers 3) Making temperature and light sensors 4) Controlling simple robots or motors.
- (g) 1) Motion — controls the movement and direction of a sprite. 2) Sound — controls playing sounds and music. 3) Events — triggers scripts to start, e.g., when a flag is clicked. 4) Control — manages the flow of the program using loops and conditions.
- (h) The stages are: 1) Planning 2) Analysis 3) Design 4) Development/Implementation 5) Testing 6) Deployment 7) Maintenance.
- (i) A web browser is software used to access, retrieve, and display content (webpages) on the World Wide Web. Two popular web browsers are Google Chrome and Mozilla Firefox.
- (j) Advantages of e-mail include fast and free communication over long distances, the ability to send attachments (documents, images), and easy record-keeping of conversations. Examples: Gmail and Outlook.

Group 'C' — Answers

- (a)
- 1) Primary memory (RAM/ROM) is directly accessible by the CPU, while secondary memory (hard disk, pen drive) is accessed indirectly through input/output channels. 2) Primary memory is generally volatile (RAM loses data when

power is off), while secondary memory is non-volatile and retains data permanently. 3) Primary memory has smaller storage capacity but faster access speed, while secondary memory has larger capacity but slower access speed. 4) Primary memory is more expensive per unit of storage compared to secondary memory.

(b)

A search engine is a software system, such as Google, designed to search for and locate information, websites, and files on the internet based on keywords entered by a user.

Pros: 1) Provides quick access to a vast amount of information. 2) Helps in research, learning, and finding services or products easily.

Cons: 1) Search results may include unreliable or biased information. 2) Overuse can reduce critical thinking as users may accept the first result without verification.

(c)

i) $110101 = 53$, $101011 = 43$. Sum = $53 + 43 = 96 \rightarrow (1100000)_2$

ii) $111010 = 58$, $011011 = 27$. Difference = $58 - 27 = 31 \rightarrow (011111)_2$

iii) $2A_3 \rightarrow 2=0010$, $A=1010$, $3=0011 \rightarrow (001010100011)_2$

iv) $450 \div 8 = 56 \text{ rem } 2$; $56 \div 8 = 7 \text{ rem } 0$; $7 \div 8 = 0 \text{ rem } 7 \rightarrow$ reading remainders bottom-up $\rightarrow (702)_8$

(d)

```
<html>
<body>
  <font color="green"><h2>My Favourite Subjects</h2></font>
  
  <ol>
    <li>Computer Science</li>
    <li>Mathematics</li>
    <li>Science</li>
  </ol>
  <p>Select your hobbies:</p>
  <input type="checkbox" name="hobby" value="Reading"> Reading<br>
  <input type="checkbox" name="hobby" value="Coding"> Coding<br>
</body>
</html>
```

(e)

Arduino UNO is an open-source microcontroller board based on the ATmega328P chip, which can be programmed to read inputs (from sensors, switches) and control outputs (LEDs, motors), commonly used for learning electronics and building interactive projects.

Applications: It is used in home automation, robotics, weather stations, obstacle-avoiding robots, and various DIY electronic projects, and is widely used in schools for teaching embedded programming.

Symbol No. _____

Sub code '3031'

Practice Question — Set 3

First Terminal Examination - 2083

Subject: Optional Computer Science

Class: IX Group 'A'

Full Marks: 50 Time: 2 hrs

Group 'A'

Rewrite the correct option in your answer sheet. [10 × 1 = 10]

1. Which unit of time is equal to one thousandth of a second?

- (a) Millisecond (b) Microsecond (c) Nanosecond (d) Picosecond

2. Which part of the computer temporarily stores data and instructions currently being processed?

- (a) Hard disk (b) RAM (c) ROM (d) Power supply unit

3. Which of the following is non-volatile and contains the startup instructions for a computer?

- (a) RAM (b) BIOS/ROM (c) Cache memory (d) Register

4. In the hexadecimal number system, what does the symbol 'E' represent?

- (a) 13 (b) 14 (c) 15 (d) 16

5. Which HTML tag is used to create a numbered (ordered) list?

- (a) (b) (c) (d) <list>

6. Which HTML tag is used to insert a line break in text?

- (a) <break> (b) <lb> (c)
 (d) <hr>

7. Which type of block in Scratch controls the sound played by a sprite?

- (a) Sound Block (b) Motion Block (c) Event Block (d) Variable Block

8. Which of the following best describes Scratch?

- (a) A text-based programming language (b) A block-based visual programming language (c) A database management tool (d) An operating system

9. Which protocol is primarily used to access websites on the internet?

- (a) SMTP (b) FTP (c) HTTP (d) POP3

10. What is a person who regularly writes and updates a blog called?

- (a) Editor (b) Blogger (c) Publisher (d) Webmaster

Group 'B'

11. Write short answers to the following questions. [10 × 2 = 20]

(a) Why is a computer called a fast device?

(b) What is the role of a hard disk in a computer system?

(c) Differentiate between system software and application software with an example each.

(d) What is the binary number system? Write down the symbols used in the binary number system.

(e) Write any two disadvantages of block-based programming.

(f) Write any four application areas of Scratch programming.

(g) Explain any four sensors/components used with micro:bit.

(h) List the stages of WDLC (Website Development Life Cycle).

(i) What are the uses of social media? Write any two popular social media platforms.

(j) Write down the advantages of blogging with an example.

Group 'C'

12. Write long answers to the following questions. [4 × 5 = 20]

- a) Differentiate between ROM and RAM. [4]
- b) Define social media. Write down its pros and cons. [4]
- c) Calculate the following as indicated: [4]
 - i) $(101110)_2 + (110101)_2$
 - ii) $(110000)_2 - (011101)_2$
 - iii) $(9F)_{16}$ into $(?)_2$
 - iv) $(300)_{10}$ into $(?)_8$
- d) Write an HTML program to create a web page including the following tags: [4]
 - i) Table
 - ii) Hyperlink
 - iii) Marquee
 - iv) Dropdown list (select)
- e) Define IoT (Internet of Things) with its applications. [4]

— **Best of Luck** —

Solutions — Practice Set 3

Group 'A' — Answer Key

1. Which unit of time is equal to one thousandth of a second? → **(a) Millisecond**
2. Which part of the computer temporarily stores data and instructions currently being processed? → **(b) RAM**
3. Which of the following is non-volatile and contains the startup instructions for a computer? → **(b) BIOS/ROM**
4. In the hexadecimal number system, what does the symbol 'E' represent? → **(b) 14**
5. Which HTML tag is used to create a numbered (ordered) list? → **(b) **
6. Which HTML tag is used to insert a line break in text? → **(c)
**
7. Which type of block in Scratch controls the sound played by a sprite? → **(a) Sound Block**
8. Which of the following best describes Scratch? → **(b) A block-based visual programming language**
9. Which protocol is primarily used to access websites on the internet? → **(c) HTTP**
10. What is a person who regularly writes and updates a blog called? → **(b) Blogger**

Group 'B' — Answers

- (a)** A computer is called a fast device because it can perform millions of calculations and instructions within a fraction of a second, far quicker than any human being.
- (b)** A hard disk is a secondary storage device that permanently stores the operating system, software programs, and user data, retaining information even when the computer is switched off.
- (c)** System software manages and controls the computer's hardware and provides a platform for other software to run, e.g., Windows OS, whereas application software is designed to help users perform specific tasks, e.g., MS Word.
- (d)** The binary number system is a base-2 number system that uses only two symbols to represent all values: 0 and 1.
- (e)** 1) It can become difficult to manage as programs grow large and complex, since blocks take up more screen space than text code. 2) It does not fully prepare learners for real-world/professional text-based programming languages.
- (f)** 1) Creating simple animations and stories 2) Building interactive games 3) Teaching basic programming logic to beginners 4) Making simulations for school science projects.
- (g)** 1) LED display — a 5x5 grid of LEDs used to show text, numbers, and images. 2) Buttons (A and B) — used to trigger actions in a program. 3) Accelerometer — detects motion, shake, and tilt. 4) Compass/magnetometer — detects direction and magnetic fields.
- (h)** The stages of WDLC are: 1) Planning 2) Analysis 3) Design 4) Development 5) Testing 6) Deployment 7) Maintenance.
- (i)** Social media is used for connecting with people, sharing photos/videos and opinions, promoting businesses, and spreading awareness or news quickly. Two popular platforms are Facebook and Instagram.
- (j)** Blogging allows a person to share knowledge, experiences, and opinions with a global audience, build a personal or professional reputation, and even earn income through advertisements. Example: a student blog about study tips on Blogger.

Group 'C' — Answers

- (a)**
- 1) ROM (Read Only Memory) is non-volatile and retains data even when power is off, while RAM (Random Access Memory) is volatile and loses data when power is off. 2) ROM mainly stores permanent startup instructions (firmware/BIOS), while RAM temporarily stores data and instructions currently being processed. 3) Data in ROM

generally cannot be modified easily, while data in RAM can be read from and written to repeatedly. 4) ROM is usually smaller in capacity, while RAM is typically larger and directly affects a computer's multitasking performance.

(b)

Social media refers to websites and applications that enable users to create and share content, and to participate in social networking with other people online.

Pros: 1) Makes communication and staying in touch with others fast and easy. 2) Provides a platform for learning, business promotion, and spreading awareness.

Cons: 1) Can lead to addiction and reduced productivity if overused. 2) Risk of exposure to misinformation, cyberbullying, and privacy issues.

(c)

i) $101110 = 46$, $110101 = 53$. Sum = $46 + 53 = 99 \rightarrow (1100011)_2$

ii) $110000 = 48$, $011101 = 29$. Difference = $48 - 29 = 19 \rightarrow (010011)_2$

iii) $9F \rightarrow 9=1001$, $F=1111 \rightarrow (10011111)_2$

iv) $300 \div 8 = 37 \text{ rem } 4$; $37 \div 8 = 4 \text{ rem } 5$; $4 \div 8 = 0 \text{ rem } 4 \rightarrow$ reading remainders bottom-up $\rightarrow (454)_8$

(d)

```
<html>
<body>
  <marquee>My School Timetable</marquee>
  <table border="1">
    <tr><th>Period</th><th>Subject</th></tr>
    <tr><td>1</td><td>Computer Science</td></tr>
    <tr><td>2</td><td>Mathematics</td></tr>
  </table><br>
  <a href="https://www.wikipedia.org">Visit Wikipedia</a><br><br>
  <label>Select your class:</label>
  <select name="class">
    <option value="9">Class 9</option>
    <option value="10">Class 10</option>
  </select>
</body>
</html>
```

(e)

IoT (Internet of Things) refers to a network of physical devices, vehicles, appliances, and other objects embedded with sensors, software, and connectivity, which allows them to collect and exchange data over the internet without human intervention.

Applications: It is used in smart homes (automated lighting and security), smart agriculture (soil moisture and irrigation monitoring), healthcare (wearable fitness/health trackers), and smart cities (traffic and waste management systems).